

LEARN PYTHON PROGRAMMING – BEGINNER TO MASTER

Section: 5. Conditional Statements

Lecture: 42. Nested if and elif Statements

Section 1: Lecture Summary

This lecture covers the advanced use of conditional statements in Python, focusing on **nested if statements**, **elif**, and the **if-elif ladder**. It begins with a quick revision of basic `if` and `else` constructs and introduces nested if, which means placing an `if` or `else` block inside another `if` or `else` block. The lecture uses the example of temperature checking — determining if a temperature is normal, cold, or hot — to illustrate nested conditions. Then, it explains the `elif` statement, which simplifies nested `if` inside `else` constructs by combining them, making the code more readable. Lastly, it introduces the concept of the **if-elif ladder**, where multiple `elif` conditions create a layered decision structure allowing several conditional checks before a final `else`. The lecture closes by noting that `elif` ladders are common in programming for handling multiple conditions cleanly, followed by mention of upcoming practice challenges.

Section 2: Key Concepts and Explanations

- Conditional Statements in Python:

- Structure:

block of statements

block of statements

```
if condition:  
    else:
```

- Indentation defines blocks belonging to each part.

- Nested If Statements:

- You can place an `if` or `else` inside another `if` or `else` block.
- Useful when decisions depend on multiple conditions.
- Example:
 - Inside `else` block, if another condition needs checking, write another if-else.
- General form:

some code

some code

some code

```
if condition1:  
    else:  
        if condition2:  
            else:
```

- Both `if` inside `if` or `if` inside `else` are valid nesting.

- Elif Statement:

- Combines `else` and `if` into a single keyword `elif`.
- Simplifies nested `if` inside `else` constructions.
- Use `elif` to check multiple exclusive conditions sequentially.
- Syntax:

code

code

code

```
if condition1:  
    elif condition2:  
    else:
```

- Helps avoid deep nesting and improves readability.

- If-Elif Ladder:

- A sequence of `if` followed by multiple `elif` and ended with an `else`.
- Each condition is checked in order until one evaluates True; corresponding block executes.
- Useful when multiple conditions need separate handling.
- Example structure:

code

code

code

code

```
if condition1:  
    elif condition2:  
    elif condition3:  
    else:
```

- Each "layer" allows for a clear, ordered decision process.
-
- **Example Scenario: Temperature Check**
 - Three outcomes based on temperature input:
 - Exactly 25: print "Normal"
 - Less than 25: print "Cold"
 - Greater than 25: print "Hot"
 - Can be implemented using nested if-else or using if-elif-else ladder.

Section 3: Example Code and Use Cases

1. Nested If Example:

```
temp = float(input("Enter Temperature: "))

if temp == 25:
    print("Normal")
else:
    if temp < 25:
        print("Cold")
    else:
        print("Hot")
```

- Explanation:

Takes temperature input as a float. First checks if temp equals 25 (normal). If not, inside the else block another if checks if temp is less than 25 (cold), else it prints hot. This demonstrates nested `if` inside `else`.

2. Elif Example:

```
temp = float(input("Enter Temperature: "))

if temp == 25:
    print("Normal")
elif temp < 25:
    print("Cold")
else:
    print("Hot")
```

- Explanation:

This code uses `elif` to combine the nested structure into a cleaner, more readable form with three condition checks: equal to 25, less than 25, and else (greater than 25).

3. If-Elif Ladder Example (general form):

action 1

action 2

action 3

default action

```
if condition1:  
elif condition2:  
elif condition3:  
else:
```

- Explanation:

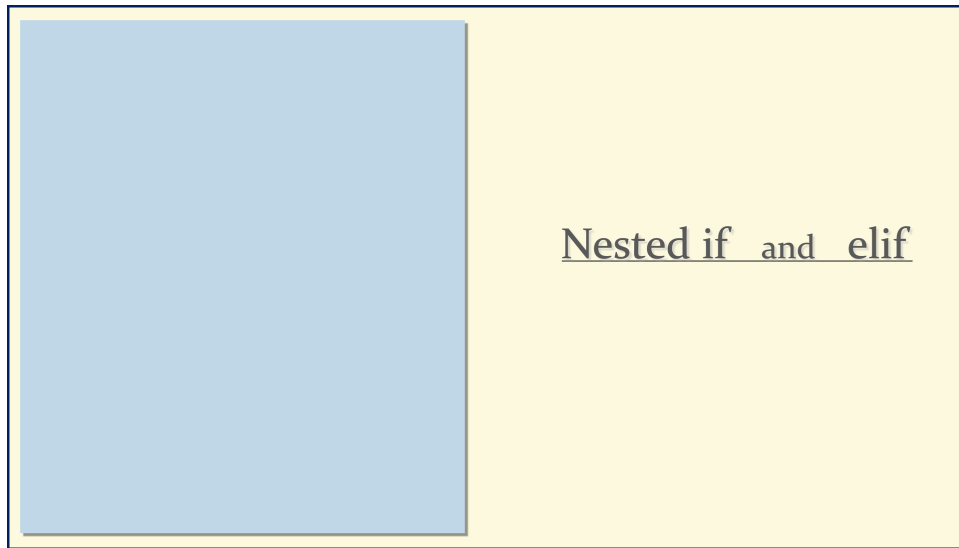
Demonstrates layered checking of multiple conditions leading to clear branching logic; useful for multiple decisions.

Section 4: Key Takeaways

- Nested `if` statements allow embedding conditionals inside others, useful but can lead to complex code if overused.
- `elif` simplifies nested `else` + `if` scenarios by combining them into one line, enhancing code readability.
- The `if-elif ladder` supports checking multiple conditions sequentially before a final fallback (`else`).
- When multiple exclusive conditions exist, use `if-elif-else` rather than multiple nested `if-else` structures.
- The temperature check example illustrates the practical use of nested `if` and `elif` ladders.
- Proper indentation is crucial for defining blocks in nested conditions.
- `elif` ladders are among the most common patterns to express multiple conditional branches in Python programs.

Lecture in Slides

Please Note: This document presents a curated selection of slides from the lecture; others have been excluded for conciseness.



Nested if and elif

Nested if and elif

- Nested if

Nested if and elif

- Nested if
- elif

Nested if and elif

- Nested if
- elif
- if elif Ladder

Nested if and elif

- Nested if

Nested if and elif

if <condition> :

else :

Nested if and elif

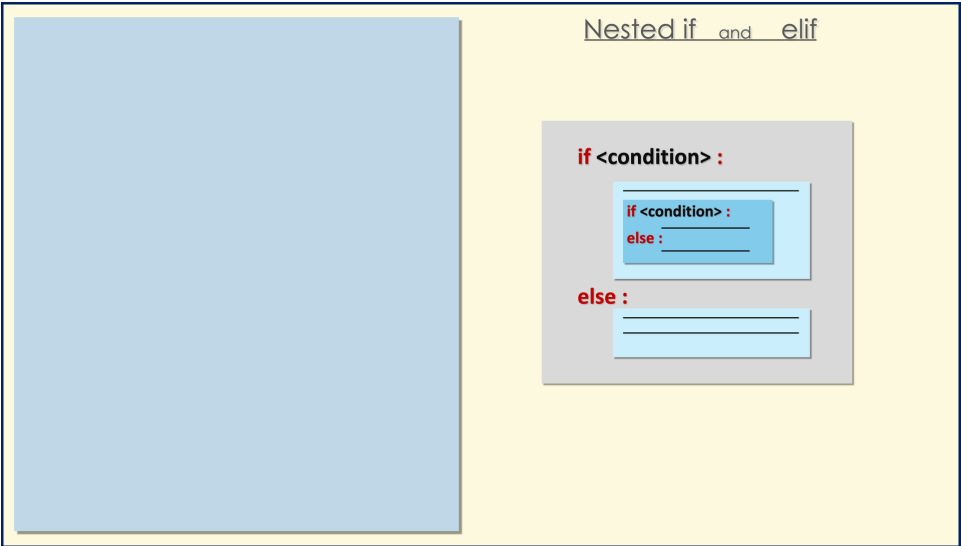
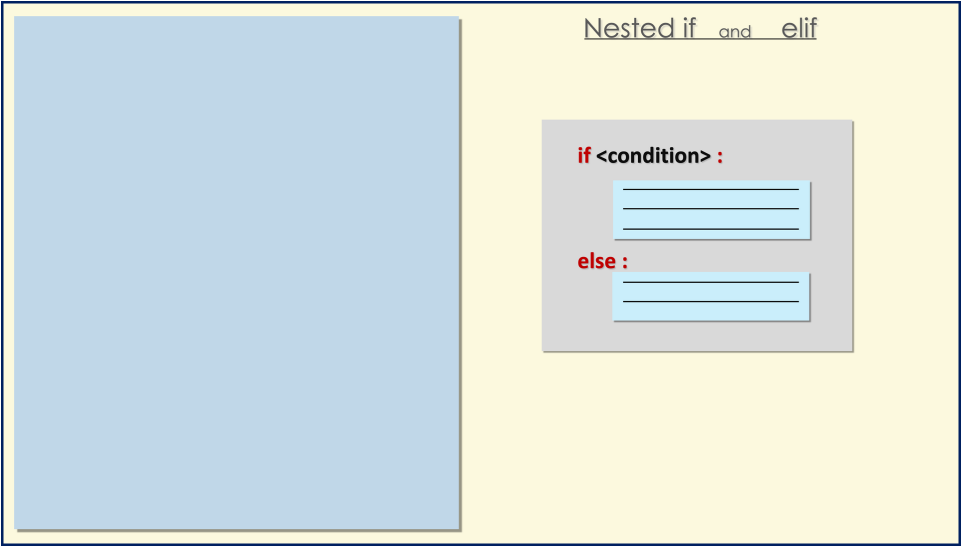
if <condition> :

else :

Nested if and elif

if <condition> :

else :



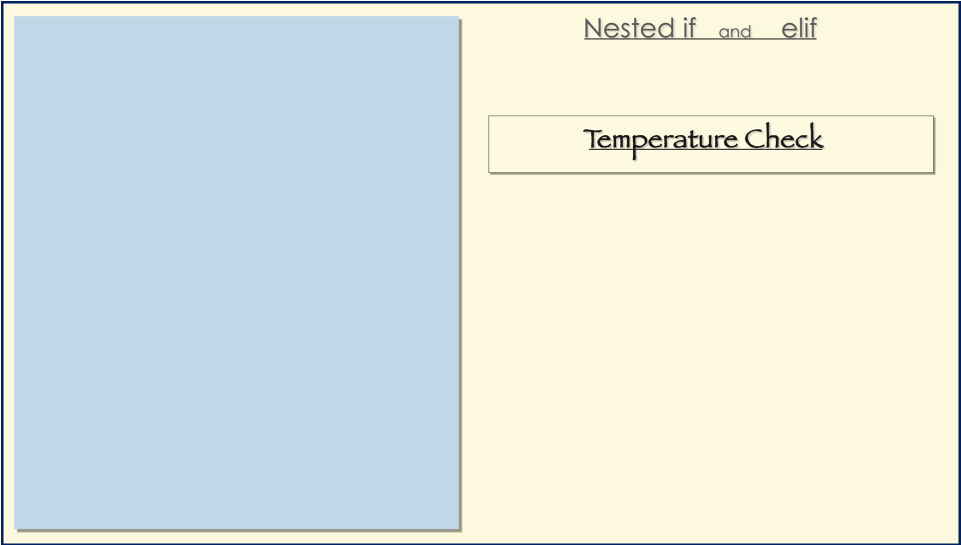
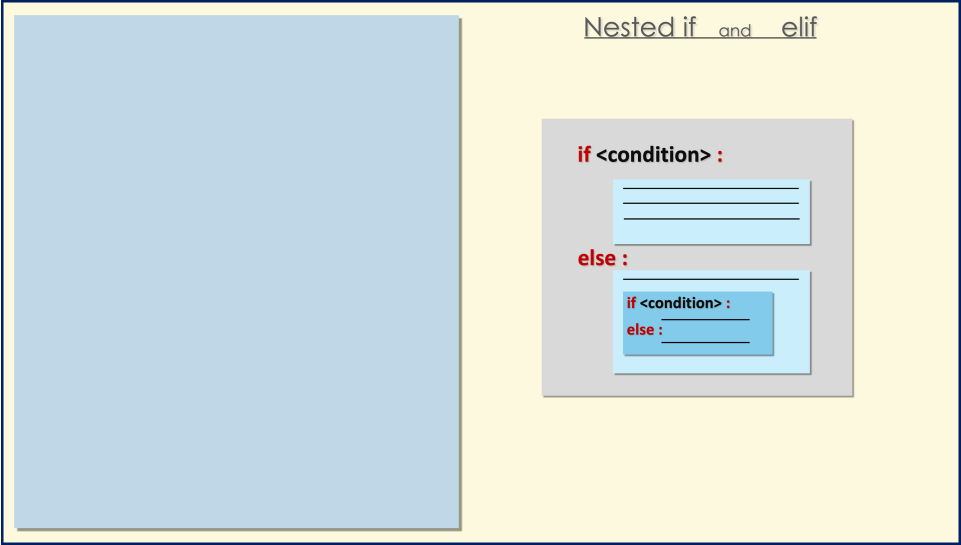
Nested if and elif

if <condition> :

if <condition> :
else : _____

else :

if <condition> :
else : _____



Nested if and elif

Temperature Check

`temp = 25`

Nested if and elif

Temperature Check

`temp = 25` `print(' Normal')`

Nested if and elif

Temperature Check

temp = 25 print(' Normal')

temp < 25

Nested if and elif

Temperature Check

temp = 25 print(' Normal')

temp < 25 print(' Cold')

Nested if and elif

Temperature Check

temp = 25 **print(' Normal')**

temp < 25 **print(' Cold')**

temp > 25

Nested if and elif

Temperature Check

temp = 25 **print(' Normal')**

temp < 25 **print(' Cold')**

temp > 25 **print(' Hot')**

Nested if and elif

Temperature Check

```
temp = 25    print(' Normal')
temp < 25    print(' Cold')
temp > 25    print(' Hot')
```

Nested if and elif

Temperature Check

```
temp = 25    print(' Normal')
temp < 25    print(' Cold')
temp > 25    print(' Hot')
```

temp

Nested if and elif

Temperature Check

```
temp = 25    print(' Normal')  
temp < 25    print(' Cold')  
temp > 25    print(' Hot')
```

```
temp = float(input( 'Enter Temperature' ))
```

Nested if and elif

Temperature Check

```
temp = 25    print(' Normal')
temp < 25    print(' Cold')
temp > 25    print(' Hot')
```

```
temp = float(input( 'Enter Temperature' ))
```

```
if temp == 25 :
```

Nested if and elif

Temperature Check

```
temp = 25    print(' Normal')
temp < 25    print(' Cold')
temp > 25    print(' Hot')
```

```
temp = float(input( 'Enter Temperature' ))
```

```
if temp == 25 :
```

```
    print( 'Normal' )
```

Nested if and elif

Temperature Check

```
temp = 25    print(' Normal')
temp < 25    print(' Cold')
temp > 25    print(' Hot')
```

```
temp = float(input( 'Enter Temperature' ))
```

```
if temp == 25 :
    print( 'Normal' )
else :
```

Nested if and elif

Temperature Check

```
temp = 25    print(' Normal')
temp < 25    print(' Cold')
temp > 25    print(' Hot')
```

```
temp = float(input( 'Enter Temperature' ))
```

```
if temp == 25 :
    print( 'Normal' )
else :
    if temp < 25 :
```

Nested if and elif

Temperature Check

```
temp = 25    print(' Normal')
temp < 25    print(' Cold')
temp > 25    print(' Hot')
```

```
temp = float(input( 'Enter Temperature' ))
```

```
if temp == 25 :
    print( 'Normal' )
else :
    if temp < 25 :
        print( 'Cold' )
```

Nested if and elif

Temperature Check

```
temp = 25    print(' Normal')
temp < 25    print(' Cold')
temp > 25    print(' Hot')
```

```
temp = float(input( 'Enter Temperature' ))
```

```
if temp == 25 :
    print( 'Normal' )
else :
    if temp < 25 :
        print( 'Cold' )
    else :
```

Nested if and elif

Temperature Check

```
temp = 25    print(' Normal')
temp < 25    print(' Cold')
temp > 25    print(' Hot')
```

```
temp = float(input( 'Enter Temperature' ))

if temp == 25 :
    print( 'Normal' )
else :
    if temp < 25 :
        print( 'Cold' )
    else :
        print( 'Hot' )
```

Nested if and elif

Temperature Check

```
temp = 25    print(' Normal')
temp < 25    print(' Cold')
temp > 25    print(' Hot')
```

```
temp = float(input( 'Enter Temperature' ))

if temp == 25 :
    print( 'Normal' )
else :
    if temp < 25 :
        print( 'Cold' )
    else :
        print( 'Hot' )
```

Nested if and elif

Temperature Check

```
temp = 25    print(' Normal')
temp < 25    print(' Cold')
temp > 25    print(' Hot')
```

```
temp = float(input( 'Enter Temperature' ))
```

```
if temp == 25 :
    print( 'Normal' )
elif temp < 25 :
    print( 'Cold' )
else :
    print( 'Hot' )
```

Nested if and elif

Temperature Check

```
temp = 25    print(' Normal')
temp < 25    print(' Cold')
temp > 25    print(' Hot')
```

```
temp = float(input( 'Enter Temperature' ))
```

```
if temp == 25 :
    print( 'Normal' )
elif temp < 25 :
    print( 'Cold' )
else :
    print( 'Hot' )
```

Nested if_and _elif

- if _elif Ladder

Nested if and elif

if elif Ladder

Nested if and elif

if elif Ladder

if <condition> :

Nested if and elif

if elif Ladder

if <condition> :

elif <condition> :

Nested if and elif

if elif Ladder

if <condition> :

elif <condition> :

elif <condition> :

Nested if and elif

if elif Ladder

```
if <condition> :  
    _____  
elif <condition> :  
    _____  
elif <condition> :  
    _____  
elif <condition> :  
    _____
```

Nested if and elif

if elif Ladder

```
if <condition> :  
    _____  
elif <condition> :  
    _____  
elif <condition> :  
    _____  
elif <condition> :  
    _____  
else :  
    _____
```

